

Program : Diploma in Artificial Intelligence & Machine Learning	
Course Code : 4342	Course Title: Machine Learning and Neural Network
Semester : 4	Credits: 4
Course Category: Program Core	
Periods per week: 4 (L:4 T:0 P:0)	Periods per semester: 60

Course Objectives:

- Students will be able to get basic knowledge on the techniques to build an intellectual machine and apply machine learning algorithms to solve problems. This course also summarizes the basic concepts of artificial neural networks.

Course Prerequisites:

Topic	Course name	Semester
Mathematics	Engineering Mathematics I & II	1 & 2
AI techniques	Artificial Intelligence	3

Course Outcomes

On completion of the course, the students will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Demonstrate supervised machine learning techniques	16	Applying
CO2	Demonstrate clustering and dimensionality reduction techniques	14	Applying
CO3	Demonstrate evaluation metrics for machine learning models	12	Applying
CO4	Demonstrate the concept of Artificial Neural Networks	16	Applying
	Series Test	2	

CO-PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Demonstrate supervised machine learning techniques		
M1.01	Explain the Machine Learning concepts.	1	Understanding
M1.02	Describe the process involved in machine learning techniques.	2	Understanding
M1.03	Describe Supervised Learning	2	Understanding
M1.04	Demonstrate Classification techniques	4	Applying
M1.05	Demonstrate Regression techniques	4	Applying
M1.06	Describe Data Preprocessing	2	Understanding
Contents: Machine Learning - Definition -Applications - Processes involved in Machine Learning, Machine Learning Techniques-Supervised Learning, Unsupervised Learning and Reinforcement Learning, Data preprocessing – Normalization, Missing value handling. Supervised Learning- Classification - Learning a Class from Examples, Logistic Regression - K-Nearest Neighbours - Support Vector Machine - Naive Bayes - Decision Tree - Random Forest Regression- Linear Regression -Multi Linear Regression			
CO2	Demonstrate Clustering & Dimensionality reduction techniques		

M2.01	Describe types of Clustering	2	Understanding
M2.02	Demonstrate Clustering techniques	5	Applying
M2.03	Illustrate Dimensionality Reduction techniques	7	Applying
	Series Test – I	1	
Contents: Clustering - Discovering clusters - Types of Clustering: Hierarchical, Agglomerative Clustering, Divisive clustering, Partitional Clustering - K-means clustering, Mean Shift clustering Dimensionality reduction – Principal Component Analysis, Factor Analysis, Multidimensional scaling, Linear Discriminant Analysis.			
CO3	Demonstrate evaluation metrics for machine learning models		
M3.01	Summarize Reinforcement learning	4	Understanding
M3.02	Describe Q learning	2	Understanding
M3.03	Demonstrate evaluation metrics for Regression	2	Applying
M3.04	Demonstrate evaluation metrics for Classification	2	Applying
M3.05	Demonstrate evaluation metrics Clustering	2	Applying
Contents: Reinforcement learning: Key features- types - Markov Decision Process-Q learning Evaluation Measures: Regression: Mean Absolute Error, Mean Squared Error, Root Mean Squared Error, R-squared. Classification: Accuracy, confusion matrix, precision, recall, F-Score, ROC-Curve. Clustering: Sum of Squared Error, Silhouette Coefficient, Dunn's Index, Ensemble methods, Bootstrapping, Cross Validation.			
CO4	Demonstrate the concept of Artificial Neural Networks		
M4.01	Summarize the concepts of Artificial Neural Network	3	Understanding
M4.02	Explain Perceptron, Training a perceptron, Multilayer perceptron	3	Understanding
M4.03	Demonstrate Back propagation Algorithm.	3	Applying

M4.04	Discuss recurrent networks	3	Understanding
M4.05	Describe Loss function, Activation function	2	Understanding
M4.06	Explain Batch normalization	2	Understanding
	Series Test – II	1	
Contents: Artificial Neural Network: Neural network representation-Structure-Advantages, Perceptron, Training a perceptron, Multilayer perceptron, Back-propagation Algorithm. Recurrent Neural Networks. Loss function, Activation function, Batch normalization.			

Text / Reference:

T/R	Book Title / Author
T1	Kevin P. Murphy. Machine Learning: A Probabilistic Perspective, MIT Press 2012.
T2	Ethem Alpaydin, Introduction to Machine Learning, 2nd edition, MIT Press 2010.
R1	Mitchell M., T., Machine Learning, McGraw Hill (1997) 1stEdition.
R2	Michie D., Spiegelhalter J. D., Taylor C. C., Campbell, J., Machine Learning, Neural and Statistical Classification. Overseas Press (1994)

Online Resources:

Sl. No.	Website Link
1	https://www.javatpoint.com/machine-learning-models
2	Introduction to Machine Learning - Course (nptel.ac.in)